

Contribution analysis

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[The theory-fit paper](#) takes rival published theories and asks which the material supports. [The realist paper](#) starts with no theory and generates mechanisms. This one starts with the theory the programme already wrote down, and asks what the evidence does to it.

Work in progress. A design, tried twice on a select committee's evidence about a proposed research funding agency. The first run failed. The second produced an annotated theory of change, and a reviewer checked its write-up against every document. Both are under *Two runs on a select committee's evidence*.

In short

Start from the theory of change the programme already wrote down, and ask what the evidence does to each link of it.

- **The theory of change is the workflow.** Each link becomes a proposition with its two ends, and one coding pass goes to each.
- **A separate rival pass.** One more coding pass over the same sources names the whole-outcome alternatives. It is not a prong of the link pass, so a rival to the outcome can never be counted as evidence for a link.
- **Who is speaking is a column.** Coding records the speaker's stake, so a second figure reports how much of the story survives once the people delivering the programme are excluded.
- **Counting that shows the holes.** The list of links is declared in advance, so a link nobody mentioned reads as a nought rather than dropping out of the table.
- **Two judgements, kept apart.** One per link, giving an annotated theory of change; one overall, against a rubric that states in advance how the links combine. A chain is as strong

as its weakest link, so the combination rule is a minimum rather than a mean.

- **What it refuses.** No attribution fraction, ever. No counterfactual: the rival pass asks what else could explain the outcome and reads what people say about it, which is weaker than a comparison group and stronger than not asking.
- **What the second run showed.** Five of six links came back well supported, including the one that most deserved doubt. A reviewer who read all 114 documents found 18 of the write-up's 24 checkable sentences sound and six wrong.

See also: [Rubicon](#); [Theory-based evaluation](#); [A simple measure of the goodness of fit of a causal theory to a text corpus](#); [Task 3 — Analysing data, Answering questions](#).

Intended audience: evaluators who have been asked to show that a programme contributed to a change, who have interviews and reports rather than a counterfactual, and who have to satisfy a reader entitled to be sceptical.

What evaluations actually want contribution analysis to answer

Worth being blunt about, because the gap between what commissioners ask and what the method delivers is where most contribution analyses go wrong.

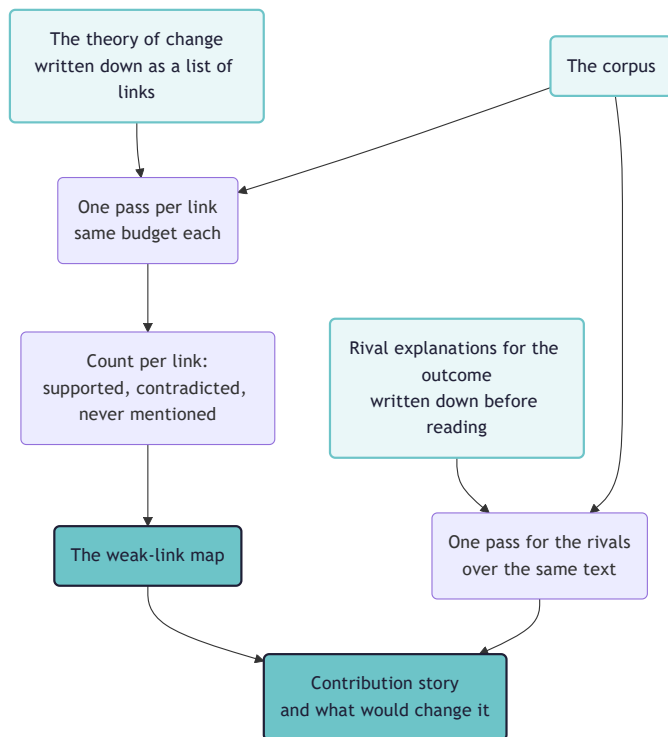
Five questions arrive dressed as one.

- **"Did our programme cause this change?"** The question as asked. Contribution analysis answers a narrower one: is there a credible account, given the evidence and given what else was going on, that the programme made a difference. Mayne's own language is reduced uncertainty and plausible association.
- **"How much of the change was ours?"** Refused, and the refusal is the method's design rather than a shortcoming. Contribution analysis produces no attribution fraction and cannot be made to. An evaluation that reports one has stopped doing contribution analysis. Say this at inception, in writing, because it is the expectation that curdles at the end.
- **"Would it have happened anyway?"** The real question underneath the first, and the one a sceptical reader will press. It is answerable only by taking the rival explanations as seriously as the programme's own account, which is the step most often skimmed.
- **"What should we fix?"** The question contribution analysis answers best and is asked to answer least. A theory of change broken into links, each carrying the evidence for and against it, is a map of where the programme's own account is thin.

That is more use to a programme manager than any overall verdict.

- **"Give us something that will survive audit."** Legitimate, and it is about the trail rather than the finding: what was claimed, on what evidence, decided by whom, and when relative to seeing the results.

So the useful reframing, and the one this paper builds on: **the product of a contribution analysis is not a verdict but an annotated theory of change**, showing for each link whether the material supports it, contradicts it, or never mentions it, alongside a separate account of what else could explain the outcome. The contribution story is the wrapper. The map is the thing.



Two kinds of rival, and the one that gets skipped

Contribution analysis carries two different things called rivals, and evaluations conflate them constantly.

- **A rival to a link.** The training did not raise referral accuracy; the new form did, and it arrived the same quarter. This is local, and the evidence that settles it is local too.
- **A rival to the whole outcome.** The change would have happened anyway, or another agency's programme did it, or the outcome is a measurement artefact of how the indicator was defined, or the sites that improved were the ones already improving.

Teams test the first kind, because each link invites its own alternative, and then write a contribution story as though the second kind had been dealt with. It has not. A theory of change every one of whose links is well evidenced is still consistent with the programme having contributed nothing, if the same forces that produced the outcome would have produced those intermediate steps too.

So the whole-outcome rivals get their own coding pass, written down before the material is read, searched with the same effort over the same text. This is the same discipline the theory-fit paper argues for, and here it is the difference between a contribution story and a press release.

A chain is as strong as its weakest link, and reports average instead

If a theory of change runs A to B to C to D, and the material says a great deal about A to B and D, and nothing at all about B to C, the story fails at B to C. It does not score three out of four.

Almost every contribution analysis reports as though it did, because the natural summary of a table is a total. So the combination rule has to be written into the rubric in advance, and it has to be closer to a minimum than a mean. Stating it in advance matters more here than anywhere else in this set of papers, because the temptation to switch to averaging arrives precisely when one link comes back empty.

In a Rubicon rubric the rule is **combine: worst:** the weakest criterion decides, read against the order in which the author declared the verdicts, best first. A rubric that never declared its verdicts is refused, since there the order would be an accident of writing. The rule cannot be nested, so a theory with several branches is written as one summary per branch and combined once.

Two qualifications keep that from being crude. A theory of change is rarely a single chain, so a weak link in one branch is not fatal where another branch reaches the same outcome. And a link nobody mentioned is not a link the material contradicts, which is the next problem.

Silence is not refutation, and it is not support either

Three outcomes per link, and the third is the one an evaluator cannot get any other way.

- **Supported**, with passages behind it.
- **Contradicted**, with passages behind it.
- **Never mentioned.** No passage bears on it at all.

The third only means anything because the workflow knows the full list of links in advance, so a proposition the material never touched reads as a nought in the table rather than dropping out of it. Build the table from what was found and the untouched links disappear, which is exactly the failure the method exists to prevent.

Silence has at least three readings, and the report should say which is being taken. The link is so obvious nobody remarks on it. Nobody was asked. Or it did not happen. The first two are facts about the corpus and the interview schedule; only the third is a finding about the programme. Where the corpus was gathered before the theory of change was written, which is the usual case for this kind of analysis, the safer default is that silence is uninformative and the link is unexamined rather than unsupported.

Who is speaking matters more here than elsewhere

An implementer's account of why participants responded is the implementer's theory, and it is data about the implementer. In a contribution analysis this bites harder than in the other papers, because the people most available to interview are the ones with the strongest interest in the contribution story being true.

Two columns earn their cost in every coding pass, both borrowed from the realist workflow.

- **First or second hand.** A programme officer reporting what a farmer felt is not the farmer.

- **Prompted or volunteered.** A respondent who raised a link unbidden is stronger evidence than one who agreed when it was put to them. In a corpus somebody else gathered, this is the only way to tell an interview that tested the theory from one that led the witness.

And a third that is specific to this method: **the speaker's relation to the programme**, so a figure can be recomputed over people with nothing to gain. Where the contribution story rests only on people paid by the programme, that is the finding.

The workflow

The theory of change is written into the workflow as a list of links, each with its two ends, and one coding pass goes to each link. Here is the coding step for two links of a theory about convening researchers and civil servants.

```

- id: test_link
  type: code
  inputs: [sources]
  for_each:

    - id: convening-builds-relationship
      from: convening
      to: relationship
      text: >-
        putting researchers and civil servants in a
room produces a working
        relationship that outlives the meeting

    - id: relationship-gets-evidence-used
      from: relationship
      to: evidence_used
      text: >-
        it is the relationship rather than the quality
of the evidence that
        gets the evidence used later
    spec:
      instruction: >
        PROPOSITION: {item.text}

        Find every passage bearing on that proposition,
for or against. A passage
        that describes the activity happening is not
evidence that it produced
        what the proposition claims; mark that as
background.
        columns:

      - name: stance
        type: category
        means: which way the passage cuts, taken at
face value
        values:

          - name: supports
            means: taken at face value, it makes the
proposition more likely

          - name: contradicts
            means: taken at face value, it makes the
proposition less likely

          - name: background
            means: >
              describes the activity or the setting
without bearing on whether
              the proposition holds

      - name: speaker_stake
        type: category
        means: what the speaker stands to gain from
the proposition being true
        values:

          - name: delivers_the_programme
            means: paid by, or accountable for, the
programme

          - name: receives_it
            means: a participant or intended
beneficiary

          - name: neither
            means: an observer with nothing riding on
the answer

      - name: prompted_by_interviewer
        type: category
        means: whether the speaker raised this or was
led to it
        values:

          - name: volunteered
            means: the speaker introduced it
themselves

```

```

- name: prompted
  means: the interviewer named it first and
the speaker agreed

- name: unclear
  means: the transcript does not show which

- name: certainty
  type: ordinal
  means: how sure the speaker sounds that what
they describe happened
  levels:

- name: speculative
  means: "imagining or generalising: would,
might, people tend to"

- name: reported
  means: stated as something that happened,
without saying how they know

- name: evidenced
  means: "particulars that show they were
there: an occasion, a person, what followed"
  outputs:

- name: link_evidence
  role: link_evidence
  type: quotes

```

The counting step gathers the prongs by role, so grouping by item gives a row per link, and the full list of links is known in advance, which is what lets a never-mentioned link read as a nought. A second figure computed with `where: {speaker_stake: {not: delivers_the_programme}}` says how much of the story survives without the people paid to tell it.

The rival pass is an ordinary coding step over the same sources, with the whole-outcome alternatives named in its instruction and a result of its own. It is not a prong of the link pass, on purpose: these are rivals to the outcome rather than to any one link, and mixing them would let a rival be counted as evidence about a link.

Then two judgements, and keeping them apart is the point.

- **Per link**, against a rubric with the three outcomes above, so the output is the annotated theory of change.
- **Overall**, against a rubric that states in advance how the links combine, how the rival explanations are weighed against the programme's account, and what would have to be true for the story to fail.

The artefact the workflow must produce, beyond either judgement, is the graph: the theory of change drawn with each link carrying what the evidence said, something distinct for a contradicted link, and something distinct again for a link nobody addressed. That last is the one an evaluator cannot get any other way and the one most easily lost in a picture. How Rubicon draws it is in [Theory-based evaluation](#).

Two runs on a select committee's evidence

The material is the evidence to a select committee inquiry into a proposed UK research funding agency: 114 written submissions and transcripts of oral sessions, published under the Open Parliament Licence. The theory of change had six links, running from a gap in the existing funding system, through the agency's budget and its freedom from the usual controls, to breakthroughs the existing system would not produce. Three whole-outcome rivals were written down before any reading.

The first run, on 5 September 2026, failed. Ruby, Rubicon's assistant, proposed the right workflow on her second turn and raised two objections this paper makes elsewhere: the sixth link cannot be evidenced from an inquiry held before the agency existed, and whether a point was volunteered or prompted cannot be coded for written submissions, which have no questions in them. The coding then read the material, returned 285 rows and kept none. Every row was refused by the columns the step had declared, 97 of them because the model labelled a proposition **L1: existing system gap** where the column declared **L1**. The run cost 2.90, and 26 further turns of conversation, costing 2.87, never produced a workflow that would run. None of that is about contribution analysis. The app was right to refuse rows labelled in a vocabulary nobody had agreed, and a conversation that keeps proposing the same repair has stopped learning from the refusals.

The second run, on 15 September 2026, read 14 of the 114 documents and produced the annotated theory of change. Five links came back well supported and one partially: that freedom from the usual peer review and reporting requirements is necessary for the agency to work. Twelve of the fourteen documents bore on the first link, that the existing system does not fund high-risk research well; twelve supported it with at least moderate bearing, six contradicted it, and eight of the supporting documents came from people who were not advocating for the agency.

The sixth link is the one to read carefully. It says research funded this way produces breakthroughs the existing system would not, and it came back well supported. The passages behind it cite what DARPA delivered in the United States (the internet, stealth aircraft, GPS) and a witness's account of a British venture that had to look abroad for backing. That is argument by analogy and by anecdote about an agency that did not yet exist, which is what Ruby had warned of before the first run. Nothing in the rubric told evidence about this agency apart from argument about agencies like it, so a link no corpus of this kind could settle looks as settled as the first. The repair belongs in the rubric, as a criterion asking what kind of evidence a link rests on, and it is the most useful thing the run taught.

A blind check of the write-up. A reviewer who had not seen the run read all 114 documents and tested every sentence of the write-up that states something about the evidence. Of 24 such sentences, 18 held. Six were wrong: speakers were merged or swapped, one witness was written up as several, a risk somebody raised was written as a conclusion, and two details appeared that the documents do not contain. Four of the six faults arose in the writing alone. The other two began in the coding, where a question from an MP and a proponent's remark were filed as evidence against a link, and a passage was filed as supporting a rival. Because every sentence cites the passages behind it, each fault could be traced to the step where it arose.

Where the people go

Contribution analysis is participatory in its own literature, and a machine-assisted version must not drop that without saying so.

- **Agreeing the theory of change** is the first and most consequential participatory step. A theory of change nobody outside the evaluation team recognises produces a link-by-link map of somebody's imagination.
- **Naming the rival explanations.** These are best drawn from people with a reason to doubt the programme, and an evaluation team on its own reliably writes rivals it can defeat.
- **Reading the weak-link map together**, which is where this method earns its keep. The map is designed for a room

rather than for a report, and the useful conversation is about the links nobody mentioned.

- **Going back for more evidence**, Mayne's fifth step. A run over an existing corpus stops before it, so the thin links become the questions for the next round of fieldwork rather than a finding.

The combination rule has to be fixed in advance, and the pull to revise it arrives exactly when a link comes back empty. Revising it is allowed as long as the revision is dated and the reader can see it came after the numbers; how that is recorded is on the [open format page](#), under pre-registering the workflow.

What this cannot do

- **No attribution fraction**, ever. See above.
- **No counterfactual.** The rival pass asks what else could explain the outcome and reads what people say about it. That is weaker than a comparison group and stronger than not asking.
- **A corpus gathered before the theory of change existed cannot have had the theory put to it.** Where an

evaluation is still running, the better use of this is to produce the weak-link map first and take the thin links into the next round of fieldwork as the questions to ask.

- **Rubicon counts documents rather than people.** Contribution is claimed about a programme and evidence arrives per person, so a figure is only as good as the match between the two.

Next steps

- Add a rubric criterion asking what kind of evidence each link rests on: about this programme, by analogy with others, or

prediction. The sixth link's verdict would then say what it is.

- Settle the combination rule for a real theory of change, in writing, before any run. Which links form a chain and which form substitutable branches is a judgement about the programme, and it belongs to the people who know it.
- Test the whole thing on a live evaluation rather than an archive, so that Mayne's fifth step exists. If you have one, we would like to hear from you.